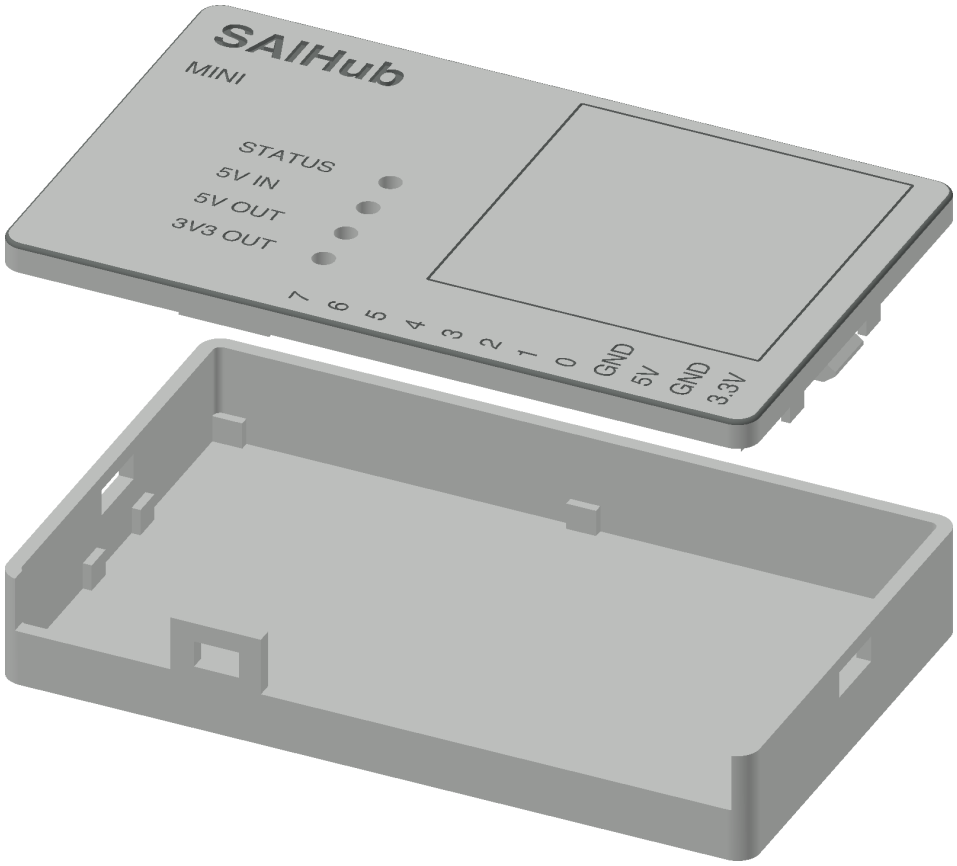
SAIHub Mini

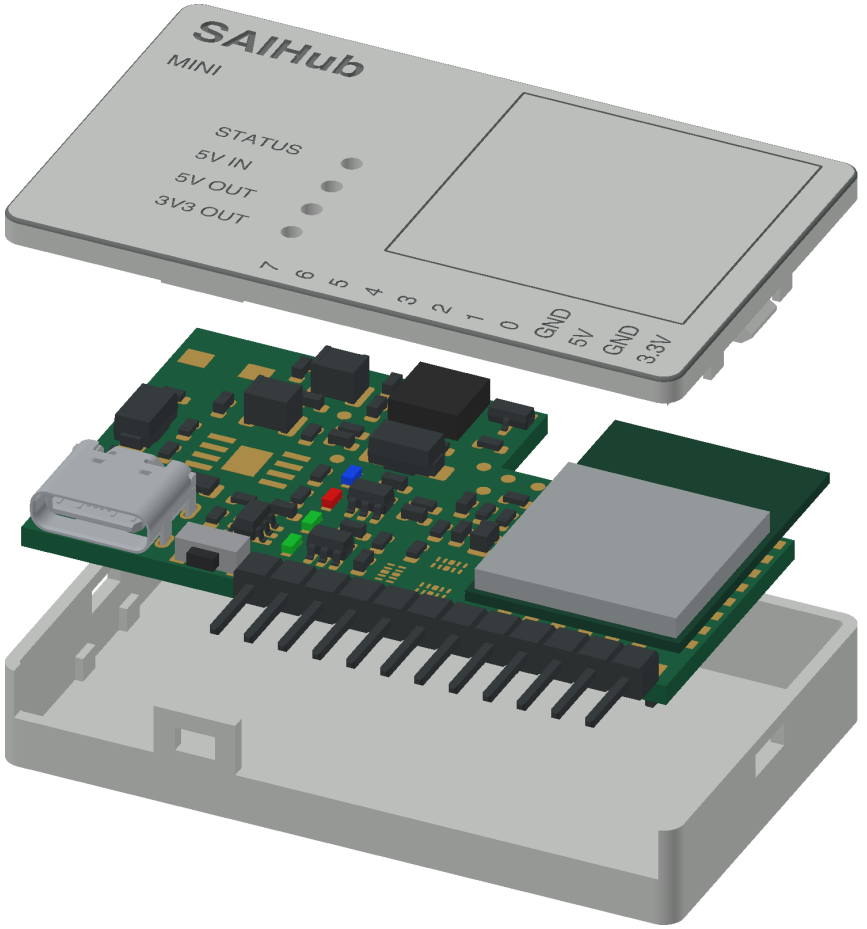
Shell and assembly

CAD-derived previews | Exploded views, not to scale

01 WITHOUT PCB



02 WITH PCB



50.4 x 30.4 x 9.5 mm overall

9.5 mm body; QR outline and lettering recessed 0.3 mm

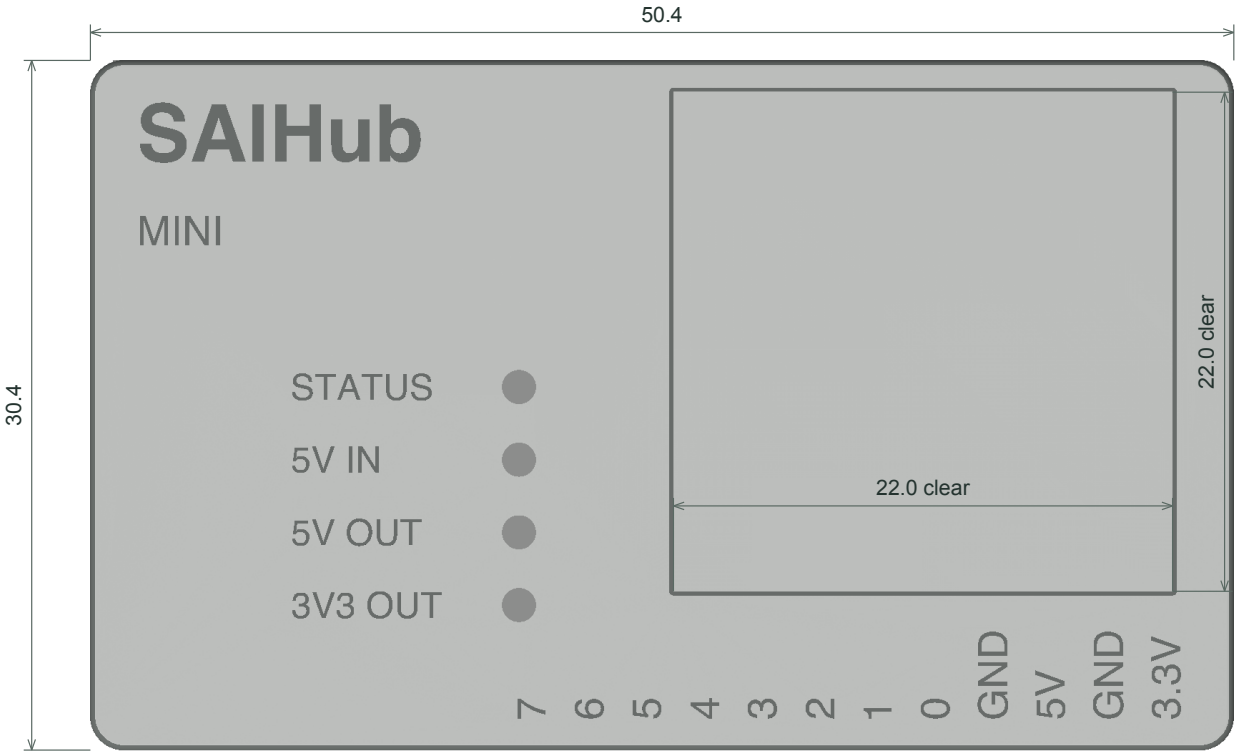
GPIO labels 0-7; four top indicators; USB and button side access.

Board: 48.0 x 28.0 x 1.6 mm, shown in assembly orientation.
Vertical separation is for illustration; XY alignment is retained.
Available models shown; button, module and LEDs are illustrative.
Shell mass excludes the PCB, components and QR sticker.

SAIHub Mini

Three-view drawing

Third-angle arrangement | Scale 3:1 at 100% print size | Shell only



TOP

DETAIL DIMENSIONS

QR outline outside: 22.36 x 22.36

Sticker bed: 22.0 x 22.0, flat

Outline width: 0.18; recess: 0.3

Text recess: 0.3 (roof remaining: 0.7)

Nominal wall / floor / roof: 1.0

Outer corner radius: R1.5

Top edge round: R0.3

Nesting overlap: 1.0; USB top strip: 1.0

Nesting clearance: 0.2 per side; 6 PCB ledges

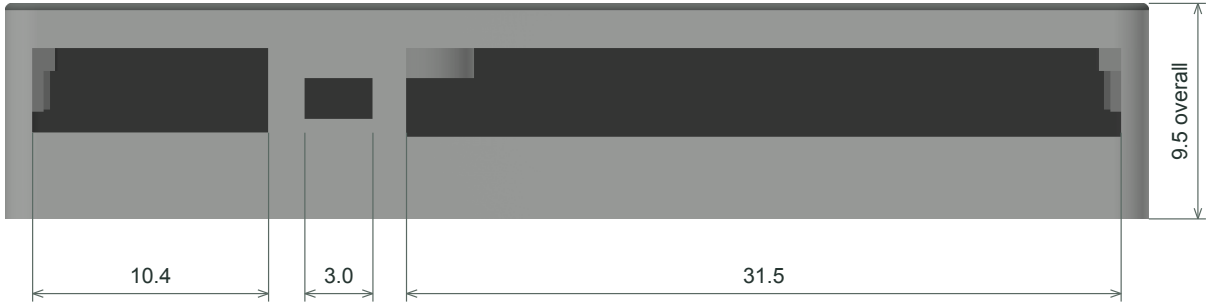
PCB clearance: 0.2 per side; roof: 4.1

2 solid ramp catches; 0.6 engagement

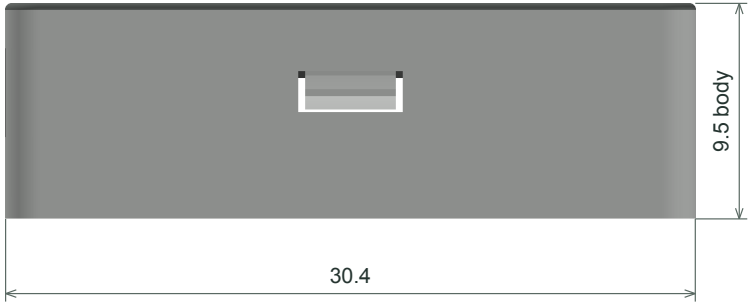
4 indicator holes: diameter 1.5; pitch 3.2

Side opening heights:
USB 3.7 / button 1.8 / header 3.9

FRONT / CONNECTOR EDGE



RIGHT SIDE



Dimensions govern. Views show external geometry; internal edges are omitted.

4.58 g

Nominal finished shell mass

Base + cover, ABS at 1.04 g/cm3

CALCULATION

Mass = CAD material volume / 1000 x ABS density

Total material volume: 4.40372 cm3

1 mm skins and walls assumed fully filled by extrusion.

The empty enclosure cavity is already excluded from volume.

PART	MATERIAL VOLUME	ESTIMATED ABS MASS
Base	2.373 cm3	2.47 g
Cover	2.031 cm3	2.11 g
TOTAL	**4.404 cm3**	**4.58 g**

PRINT ESTIMATE

Excludes supports, brim, purge, PCB, components and sticker.

Actual mass varies with filament density, line width, gaps and flow.

Do not multiply this thin-wall estimate by a general infill percentage.

Use the slicer's filament mass for the final chosen print settings.

FIT CHECKS

Valid connected solids; closed STL meshes.

No PCB overlap; USB model is taller than tested hardware.

USB model: 0.205 mm local overlap; physical fit user-verified.

Prototype-test side snaps, solder tails and repeated lid removal.

Release: gently lift each side in turn; walls flex outward.

Catches: 4.0 wide; 0.8 projection; windows: 4.6 x 1.8.

Density source: PolyLite ABS Technical Data Sheet, V5.5, 1.04 g/cm3 at 23 C.

Open manufacturer data sheet

Geometry source: hardware/mechanical/enclosure.py and reference/pcb.step. Nominal, not production-toleranced.